CHAPTER 10

SERVER SIDE SCRIPTING USING PHP

CHAPTER ESSENCE AT A GLANCE:
Develop dynamic server-side applications using PHP scripting, superglobal arrays, form handling, session tracking, and MySQL database connectivity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Server-Side Interpretation:** PHP code executes entirely within the server environment before delivering pure generated hypertext markup to the client browser, completely shielding source scripts and business logic from user inspection.

• **Dynamic Variables and Flexible Arrays:** Variables require leading currency prefixes and support dynamic type switching; arrays support both zero-indexed sequential collections and key-value associative dictionaries.

• **Form Submission Channels:** Form data submitted through the uniform resource locator query string exposes parameter values publicly, whereas data submitted through the request body remains concealed and handles extensive payloads securely.

• **Stateful Session Preservation:** Because the fundamental hypertext transfer protocol is stateless, web applications rely on server sessions and client cookies to maintain authenticated user state across sequential page visits.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Browser Request } (.\text{php}) \xrightarrow{\text{Web Server}} \text{PHP Engine} \xrightarrow{\text{Pure HTML}} \text{Rendered View}$$

Server-side interpretation and client-rendered HTML lifecycle

$$\$POST[\text{"key"}] \rightarrow \text{HTTP Body (Secure)} \quad | \quad \$GET[\text{"key"}] \rightarrow \text{URL Query (Visible)}$$

PHP superglobal array form data collection mechanisms

$$\text{mysqliConnect}() \longrightarrow \text{mysqliQuery}() \longrightarrow \text{mysqliFetch}() \longrightarrow \text{mysqliClose}()$$

Systematic MySQL database transaction workflow in PHP

$$\text{Data Density} = \frac{\text{Valid Payload Fields}}{\text{Total Form Fields}}$$

Form submission payload validation efficiency ratio

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** PHP script code must be enclosed inside <?php ... ?> tags and execute strictly on the server before sending HTML to the client.

• **Alert:** Variable names in PHP must begin with a dollar sign ($) followed by a letter or underscore, and are strictly case-sensitive.

• **Alert:** Always sanitize form inputs using htmlspecialchars() to prevent Cross-Site Scripting (XSS) vulnerabilities.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is PHP? Explain the characteristics and advantages of PHP as a server-side scripting language. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Definition):** PHP (Hypertext Preprocessor) is an open-source, interpreted, server-side scripting language designed primarily for web development. **[1 Mark]**

• **Step 2 (Key Features):** 1. Server-side execution (cross-platform). 2. Excellent MySQL database integration. 3. Fast execution. 4. Open source and free. **[1 Mark]**

• **Key Exam Result:** PHP: Open-source server-side language with native MySQL support



Server-side interpretation and client-rendered HTML lifecycle

2. How is PHP code embedded in an HTML document? Explain PHP tags and the echo statement. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Script Tags):** <?php ... ?>. **[0.5 Mark]**

• **Step 2 (echo vs print):** echo: Output construct; can take multiple parameters separated by commas (echo 'A', 'B'); has no return value; slightly faster. print: Function-like; takes only one argument; always returns 1. **[1.5 Marks]**

• **Key Exam Result:** <?php echo 'Hello World'; ?>; echo has no return value, print returns 1



Server-side interpretation and client-rendered HTML lifecycle

3. Explain control structures in PHP: if-else, switch, while, for, and foreach loops. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Control Structures):** Conditional: if, if-else, if-elseif-else, switch($val) { case 1: ... break; }. Loops: while (condition), do-while, for ($i=0; $i<n; $i++), foreach ($arr as $val) (for arrays). **[3 Marks]**

• **Key Exam Result:** if-else, switch, while, for, foreach



Server-side interpretation and client-rendered HTML lifecycle

4. Explain arrays in PHP: Indexed Arrays vs Associative Arrays with syntax examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Indexed Arrays):** Numeric integer keys: $colors = array('Red', 'Green', 'Blue'); echo $colors[0]; **[1.5 Marks]**

• **Step 2 (Associative Arrays):** Named string keys: $marks = array('Physics' => 92, 'Maths' => 98); echo $marks['Maths']; **[1.5 Marks]**

• **Key Exam Result:** Indexed (numeric keys) vs Associative (named string keys =>)



Server-side interpretation and client-rendered HTML lifecycle

5. Explain the systematic five steps to connect PHP with a MySQL database. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Five Steps):** 1. Connect: $conn = mysqliConnect('localhost', 'user', 'pass', 'schoolDb'); 2. Check error: if (!$conn) die('Connection failed: ' . mysqliConnectError()); 3. Execute query: $res = mysqliQuery($conn, $sql); 4. Fetch records: while($row = mysqliFetchAssoc($res)) { ... } 5. Close: mysqliClose($conn); **[4 Marks]**

• **Key Exam Result:** mysqliConnect --> mysqliQuery --> mysqliFetchAssoc --> mysqliClose



6. Write a PHP script to insert student form data into a MySQL table. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Insert Script):** <?phpn $conn = mysqliConnect('localhost', 'root', '', 'school');n $roll = $POST['roll']; $name = $POST['name']; $marks = $POST['marks'];n $sql = "INSERT INTO student VALUES ($roll, '$name', $marks)";n if (mysqliQuery($conn, $sql)) echo 'Record inserted successfully!';n mysqliClose($conn);n?> **[5 Marks]**

• **Key Exam Result:** PHP script inserting form data into MySQL



7. Write a PHP script to retrieve and display student records from a MySQL table in an HTML table. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Display Script):** <?phpn $conn = mysqliConnect('localhost', 'root', '', 'school');n $res = mysqliQuery($conn, 'SELECT * FROM student');n echo '<table border="1"><tr><th>Roll</th><th>Name</th></tr>';n while ($row = mysqliFetchAssoc($res)) {n echo '<tr><td>' . $row['roll'] . '</td><td>' . $row['name'] . '</td></tr>';n }n echo '</table>'; mysqliClose($conn);n?> **[5 Marks]**

• **Key Exam Result:** PHP script rendering MySQL records in HTML table

